UTTARANCHAL UNIVERSITY

(Established vide Uttaranchal University Act, 2012)
(Uttarakhand Act No. 11 of 2013)
Arcadia Grant, P.O. Chandanwari, Premnagar, Dehradun, Uttarakhand, India

Practical File

INFORMATION SECURITY AND PRIVACY LAB

PCS-354

Submitted To:

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CSE-3, 3rd Year

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Program-1

Implement the **Caesar Cypher**.

Program:

```
// A  C++ program to illustrate the Caesar Cypher Technique

#include <iostream>
using namespace std;

// This function receives text and shift and returns the encrypted text
string encrypt(string text, int s)
{
   string result = "";

   // traverse text
   for (int i = 0; i < text.length(); i++) {
      // apply transformation to each character
      // Encrypt Uppercase letters
      if (isupper(text[i]))
         result += char(int(text[i] + s - 65) % 26 + 65);

      // Encrypt Lowercase letters
      else
         result += char(int(text[i] + s - 97) % 26 + 97);
   }

   // Return the resulting string
   return result;
}

// Driver program to test the above function
int main()
{
   string text = "ATTACKATONCE";
   int s = 4;
   cout << "Text : " << text;
   cout << "\nShift: " << s;
   cout << "\nCipher: " << encrypt(text, s);
   return 0;
}
```

Output:

```
Text : ATTACKATONCE
Shift: 4
Cipher: EXXEGOEXSRGI

...Program finished with exit code 0
Press ENTER to exit console.
```

Program-2

Implement the **Affine Cypher** using the equation: **C = 3x + 12**.

Program:

```
//CPP program to illustrate Affine Cypher
#include<bits/stdc++.h>
using namespace std;

//Key values of a and b
const int a = 17;
const int b = 20;
string encryptMessage(string msg)
{
   ///CypherText initially empty
   string cypher= "";
   for (int i = 0; i < msg.length(); i++)
   {
      // Avoid space to be encrypted
      if(msg[i]!=' ')
         /* applying encryption formula ( a x + b ) mod m
         {here x is msg[i] and m is 26} and added 'A' to
         bring it in range of ascii alphabet[ 65-90 | A-Z ] */
         cypher= cypher+
                  (char) ((((a * (msg[i]-'A') ) + b) % 26) + 'A');
      else
         //else simply append space character
         cypher+= msg[i];
   }
   return cypher;
}

string decryptCypher(string cypher)
{
```

```
    string msg = "";
    int a_inv = 0;
    int flag = 0;
    //Find a^-1 (the multiplicative inverse of a
        //in the group of integers modulo m.)
    for (int i = 0; i < 26; i++)
    {
        flag = (a * i) % 26;
        //Check if (a*i)%26 == 1,
                //then i will be the multiplicative inverse of a
        if (flag == 1)
        {
            a_inv = i;
        }
    }
    for (int i = 0; i < cypher.length(); i++)
    {
        if(cipher[i]!=' ')
            /*Applying decryption formula a^-1 ( x - b ) mod m
            {here x is cipher[i] and m is 26} and added 'A'
            to bring it in range of ASCII alphabet[ 65-90 | A-Z ] */
            msg = msg +
                    (char) (((a_inv * ((cypher[i]+'A' - b)) % 26)) + 'A');
        else
            //else simply append space character
            msg += cypher[i];
    }

    return msg;
}
//Driver Program
int main(void)
```

```
{
    string msg = "AFFINE CYPHER";
   //Calling encryption function
    string cipherText = encryptMessage(msg);
    cout << "Encrypted Message is : " << cypherText<<endl;
     //Calling Decryption function
    cout << "Decrypted Message is: " << decryptCypher(cypherText);
    return 0;
}
```

Output:

```
Encrypted Message is : UBBAHK CMPJKX
Decrypted Message is: AFFINE CYPHER

...Program finished with exit code 0
Press ENTER to exit console.
```

Program-3

Implement the **Playfair Cypher** with the key entered by the user.

Program:

```
#include <bits/stdc++.h>
using namespace std;
// Function to convert the string to lowercase
void toLowerCase(string &plain) {
    int n = plain.size();
    for (int i = 0; i < n; i++) {
        if (plain[i] > 64 && plain[i] < 91)
            plain[i] += 32;
    }
}

// Function to remove all spaces in a string
void removeSpaces(string &plain) {
    int n = plain.size();
    string temp;
    for (int i = 0; i < n; i++) {
        if (plain[i] != ' ') {
            temp += plain[i];
        }
    }
    plain = temp;
}

// Function to generate the 5x5 key square
void generateKeyTable(string &key, vector<vector<char>> &keyT) {
    int n = key.size();

    keyT.resize(5, vector<char>(5, 0));
    vector<int> hash(26, 0);
```

```
int i, j, k;
for (i = 0; i < n; i++) {
    if (key[i] != 'j')
        hash[key[i] - 97] = 2;
}

hash['j' - 97] = 1;

i = 0;
j = 0;

for (k = 0; k < n; k++) {
    if (hash[key[k] - 97] == 2) {
        hash[key[k] - 97] -= 1;
        keyT[i][j] = key[k];
        j++;
        if (j == 5) {
            i++;
            j = 0;
        }
    }
}

for (k = 0; k < 26; k++) {
    if (hash[k] == 0) {
        keyT[i][j] = (char)(k + 97);
        j++;
        if (j == 5) {
            i++;
            j = 0;
        }
```

```
        }
    }
}

// Function to search for the characters
void search(vector<vector<char>> &keyT, char a, char b, vector<int> &arr) {
    int i, j;
    if (a == 'j')
        a = 'i';
    else if (b == 'j')
        b = 'i';

    for (i = 0; i < 5; i++) {
        for (j = 0; j < 5; j++) {
            if (keyT[i][j] == a) {
                arr[0] = i;
                arr[1] = j;
            }
            else if (keyT[i][j] == b) {
                arr[2] = i;
                arr[3] = j;
            }
        }
    }
}

// Function to make plaintext length even
int prepare(string &str) {
    if (str.size() % 2 != 0) {
        str += 'z';
    }
    return str.size();
```

```
}
// Encryption function
void encrypt(string &str, vector<vector<char>> &keyT) {
    int n = str.size();
    vector<int> arr(4);
    for (int i = 0; i < n; i += 2) {
        search(keyT, str[i], str[i + 1], arr);
        if (arr[0] == arr[2]) {
            str[i] = keyT[arr[0]][(arr[1] + 1) % 5];
            str[i + 1] = keyT[arr[0]][(arr[3] + 1) % 5];
        }
        else if (arr[1] == arr[3]) {
            str[i] = keyT[(arr[0] + 1) % 5][arr[1]];
            str[i + 1] = keyT[(arr[2] + 1) % 5][arr[1]];
        }
        else {
            str[i] = keyT[arr[0]][arr[3]];
            str[i + 1] = keyT[arr[2]][arr[1]];
        }
    }
}

// Playfair Cipher
void encryptByPlayfairCipher(string &str, string &key) {
    vector<vector<char>> keyT;
    removeSpaces(key);
    toLowerCase(key);
    toLowerCase(str);
    removeSpaces(str);
    prepare(str);
    generateKeyTable(key, keyT);
    encrypt(str, keyT)
```

```
}
int main() {
    string key, str;
    cout << "Enter Key: ";
    getline(cin, key);
    cout << "Enter Plain Text: ";
    getline(cin, str);
    cout << "Key text: " << key << endl;
    cout << "Plain text: " << str << endl;
    encryptByPlayfairCipher(str, key);
    cout << "Cipher text: " << str << endl;
    return 0;
}
```

Output :

```
Enter Key: Monarchy
Enter Plain Text: instruments
Key text: Monarchy
Plain text: instruments
Cipher text: gatlmzclrqtx


...Program finished with exit code 0
Press ENTER to exit console.
```

Program-4

Implement **a simple RSA algorithm** using small numbers.

Program:

```
#include <bits/stdc++.h>

using namespace std;

// Function to compute base^expo mod m
int power(int base, int expo, int m) {
    int res = 1;
    base = base % m;
    while (expo > 0) {
        if (expo & 1)
            res = (res * 1LL * base) % m;
        base = (base * 1LL * base) % m;
        expo = expo / 2;
    }
    return res;
}

// Function to find modular inverse of e modulo phi(n)
// Here we are calculating phi(n) using Hit and Trial Method
// but we can optimize it using Extended Euclidean Algorithm
int modInverse(int e, int phi) {
    for (int d = 2; d < phi; d++) {
        if ((e * d) % phi == 1)
            return d;
    }
    return -1;
}

// RSA Key Generation
```

```
void generateKeys(int &e, int &d, int &n) {
    int p = 7919;
    int q = 1009;

    n = p * q;
    int phi = (p - 1) * (q - 1);

    // Choose e, where 1 < e < phi(n) and gcd(e, phi(n)) == 1
    for (e = 2; e < phi; e++) {
        if (__gcd(e, phi) == 1)
            break;
    }

    // Compute d such that e * d ≡ 1 (mod phi(n))
    d = modInverse(e, phi);
}

// Encrypt message using public key (e, n)
int encrypt(int m, int e, int n) {
    return power(m, e, n);
}

// Decrypt message using private key (d, n)
int decrypt(int c, int d, int n) {
    return power(c, d, n);
}

int main() {
    int e, d, n;

    // Key Generation
    generateKeys(e, d, n);
```

```
    cout << "Public Key (e, n): (" << e << ", " << n << ")\n";
    cout << "Private Key (d, n): (" << d << ", " << n << ")\n";

    // Message
    int M = 123;
    cout << "Original Message: " << M << endl;

    // Encrypt the message
    int C = encrypt(M, e, n);
    cout << "Encrypted Message: " << C << endl;

    // Decrypt the message
    int decrypted = decrypt(C, d, n);
    cout << "Decrypted Message: " << decrypted << endl;

    return 0;
}
```

Output:

```
Public Key (e, n): (5, 7990271)
Private Key (d, n): (1596269, 7990271)
Original Message: 123
Encrypted Message: 3332110
Decrypted Message: 123


...Program finished with exit code 0
Press ENTER to exit console.
```

Program-5

Implement **DES encryption and decryption.**

Program:

```
#include <bits/stdc++.h>
using namespace std;

// Hex to binary
string hex2bin(string s) {
    unordered_map<char, string> mp = {
        {'0',"0000"},{'1',"0001"},{'2',"0010"},{'3',"0011"},
        {'4',"0100"},{'5',"0101"},{'6',"0110"},{'7',"0111"},
        {'8',"1000"},{'9',"1001"},{'A',"1010"},{'B',"1011"},
        {'C',"1100"},{'D',"1101"},{'E',"1110"},{'F',"1111"}
    };
    string bin = "";
    for(char c : s) bin += mp[c];
    return bin;
}

// Binary to hex
string bin2hex(string s) {
    unordered_map<string, string> mp = {
        {"0000","0"},{"0001","1"},{"0010","2"},{"0011","3"},
        {"0100","4"},{"0101","5"},{"0110","6"},{"0111","7"},
        {"1000","8"},{"1001","9"},{"1010","A"},{"1011","B"},
        {"1100","C"},{"1101","D"},{"1110","E"},{"1111","F"}
    };
    string hex = "";
    for(int i = 0; i < s.length(); i += 4) {
        string ch = s.substr(i,4);
        hex += mp[ch];
    }
```

```
    return hex;
}

// Binary to decimal
int bin2dec(string binary) {
    return stoi(binary, 0, 2);
}

// Decimal to binary (4-bit)
string dec2bin(int num) {
    string res = bitset<4>(num).to_string();
    return res;
}

// Permutation
string permute(string k, int arr[], int n) {
    string per = "";
    for(int i = 0; i < n; i++)
        per += k[arr[i] - 1];
    return per;
}

// Left shift
string shift_left(string k, int shifts) {
    while(shifts--) {
        k = k.substr(1) + k[0];
    }
    return k;
}

// XOR
string xor_op(string a, string b) {
```

```
    string ans = "";
    for(int i = 0; i < a.size(); i++) {
        ans += (a[i] == b[i]) ? '0' : '1';
    }
    return ans;
}

// Tables
int initial_perm[64] = {
58,50,42,34,26,18,10,2,60,52,44,36,28,20,12,4,
62,54,46,38,30,22,14,6,64,56,48,40,32,24,16,8,
57,49,41,33,25,17,9,1,59,51,43,35,27,19,11,3,
61,53,45,37,29,21,13,5,63,55,47,39,31,23,15,7
};

int exp_d[48] = {
32,1,2,3,4,5,4,5,6,7,8,9,8,9,10,11,
12,13,12,13,14,15,16,17,16,17,18,19,20,21,
20,21,22,23,24,25,24,25,26,27,28,29,28,29,
30,31,32,1
};

int per[32] = {
16,7,20,21,29,12,28,17,1,15,23,26,
5,18,31,10,2,8,24,14,32,27,3,9,
19,13,30,6,22,11,4,25
};

int final_perm[64] = {
40,8,48,16,56,24,64,32,39,7,47,15,55,23,63,31,
38,6,46,14,54,22,62,30,37,5,45,13,53,21,61,29,
36,4,44,12,52,20,60,28,35,3,43,11,51,19,59,27,
```

```
34,2,42,10,50,18,58,26,33,1,41,9,49,17,57,25
};

// S-box
int sbox[8][4][16] = {
    // (same as your Python S-box, omitted here for brevity — keep identical)
};

string encrypt(string pt, vector<string> rkb, vector<string> rk) {
    pt = hex2bin(pt);
    pt = permute(pt, initial_perm, 64);

    string left = pt.substr(0,32);
    string right = pt.substr(32,32);

    for(int i = 0; i < 16; i++) {
        string right_expanded = permute(right, exp_d, 48);
        string x = xor_op(right_expanded, rkb[i]);

        string sbox_str = "";
        for(int j = 0; j < 8; j++) {
            string row_bits = "" + string(1,x[j*6]) + string(1,x[j*6+5]);
            int row = bin2dec(row_bits);

            string col_bits = x.substr(j*6+1,4);
            int col = bin2dec(col_bits);

            int val = sbox[j][row][col];
            sbox_str += dec2bin(val);
        }

        sbox_str = permute(sbox_str, per, 32);
```

```
        string result = xor_op(left, sbox_str);
        left = result;

        if(i != 15) swap(left, right);

        cout << "Round " << i+1 << " "
            << bin2hex(left) << " "
            << bin2hex(right) << " "
            << rk[i] << endl;
    }

    string combine = left + right;
    string cipher = permute(combine, final_perm, 64);
    return cipher;
}

int main() {
    string pt = "123456ABCD132536";
    string key = "AABB09182736CCDD";

    key = hex2bin(key);

    int keyp[56] = {
        57,49,41,33,25,17,9,1,58,50,42,34,26,18,
        10,2,59,51,43,35,27,19,11,3,60,52,44,36,
        63,55,47,39,31,23,15,7,62,54,46,38,30,22,
        14,6,61,53,45,37,29,21,13,5,28,20,12,4
    };

    key = permute(key, keyp, 56);

    int shift_table[16] = {
```

```
    1,1,2,2,2,2,2,2,1,2,2,2,2,2,2,1
};

int key_comp[48] = {
    14,17,11,24,1,5,3,28,15,6,21,10,
    23,19,12,4,26,8,16,7,27,20,13,2,
    41,52,31,37,47,55,30,40,51,45,
    33,48,44,49,39,56,34,53,46,
    42,50,36,29,32
};

string left = key.substr(0,28);
string right = key.substr(28,28);

vector<string> rkb, rk;

for(int i = 0; i < 16; i++) {
    left = shift_left(left, shift_table[i]);
    right = shift_left(right, shift_table[i]);

    string combine = left + right;
    string round_key = permute(combine, key_comp, 48);

    rkb.push_back(round_key);
    rk.push_back(bin2hex(round_key));
}

cout << "Encryption\n";
string cipher = bin2hex(encrypt(pt, rkb, rk));
cout << "Cipher Text: " << cipher << endl;

cout << "Decryption\n";
```

```
    reverse(rkb.begin(), rkb.end());
    reverse(rk.begin(), rk.end());

    string text = bin2hex(encrypt(cipher, rkb, rk));
    cout << "Plain Text: " << text << endl;

    return 0;
}
```

Output:

```
Encryption
Round 1 18CA18AD 14A7D678 194CD072DE8C
Round 2 14A7D678 18CA18AD 4568581ABCCE
Round 3 18CA18AD 14A7D678 06EDA4ACF5B5
Round 4 14A7D678 18CA18AD DA2D032B6EE3
Round 5 18CA18AD 14A7D678 69A629FEC913
Round 6 14A7D678 18CA18AD C1948E87475E
Round 7 18CA18AD 14A7D678 708AD2DDB3C0
Round 8 14A7D678 18CA18AD 34F822F0C66D
Round 9 18CA18AD 14A7D678 84BB4473DCCC
Round 10 14A7D678 18CA18AD 02765708B5BF
Round 11 18CA18AD 14A7D678 6D5560AF7CA5
Round 12 14A7D678 18CA18AD C2C1E96A4BF3
Round 13 18CA18AD 14A7D678 99C31397C91F
Round 14 14A7D678 18CA18AD 251B8BC717D0
Round 15 18CA18AD 14A7D678 3330C5D9A36D
Round 16 18CA18AD 14A7D678 181C5D75C66D
Cipher Text: 2138A957CE231A39
Decryption
Round 1 14A7D678 18CA18AD 181C5D75C66D
Round 2 18CA18AD 14A7D678 3330C5D9A36D
Round 3 14A7D678 18CA18AD 251B8BC717D0
Round 4 18CA18AD 14A7D678 99C31397C91F
Round 5 14A7D678 18CA18AD C2C1E96A4BF3
Round 6 18CA18AD 14A7D678 6D5560AF7CA5
Round 7 14A7D678 18CA18AD 02765708B5BF
Round 8 18CA18AD 14A7D678 84BB4473DCCC
Round 9 14A7D678 18CA18AD 34F822F0C66D
Round 10 18CA18AD 14A7D678 708AD2DDB3C0
Round 11 14A7D678 18CA18AD C1948E87475E
Round 12 18CA18AD 14A7D678 69A629FEC913
Round 13 14A7D678 18CA18AD DA2D032B6EE3
Round 14 18CA18AD 14A7D678 06EDA4ACF5B5
Round 15 14A7D678 18CA18AD 4568581ABCCE
Round 16 14A7D678 18CA18AD 194CD072DE8C
Plain Text: 123456ABCD132536


...Program finished with exit code 0
Press ENTER to exit console.
```

Program-6

Implement **AES encryption and decryption.**

Program:

```
#include <iostream>
#include <string>

#include "cryptlib.h"
#include "aes.h"
#include "modes.h"
#include "filters.h"
#include "hex.h"
#include "osrng.h"

using namespace std;
using namespace CryptoPP;

int main() {
    AutoSeededRandomPool prng;

    // Key and IV
    byte key[AES::DEFAULT_KEYLENGTH];
    prng.GenerateBlock(key, sizeof(key));

    byte iv[AES::BLOCKSIZE];
    prng.GenerateBlock(iv, sizeof(iv));

    string plain = "CBC Mode Test";
    string cipher, encoded, recovered;

    // Print Key
    encoded.clear();
    StringSource(key, sizeof(key), true,
```

```
    new HexEncoder(new StringSink(encoded))
);
cout << "Key: " << encoded << endl;

// Print IV
encoded.clear();
StringSource(iv, sizeof(iv), true,
    new HexEncoder(new StringSink(encoded))
);
cout << "IV: " << encoded << endl;

// Encryption
try {
    cout << "Plain Text: " << plain << endl;

    CBC_Mode<AES>::Encryption encryptor;
    encryptor.SetKeyWithIV(key, sizeof(key), iv);

    StringSource(plain, true,
        new StreamTransformationFilter(encryptor,
            new StringSink(cipher)
        )
    );
}
catch (const Exception& e) {
    cerr << e.what() << endl;
    return 1;
}

// Print Cipher Text
encoded.clear();
StringSource(cipher, true,
```

```
        new HexEncoder(new StringSink(encoded))
    );
    cout << "Cipher Text: " << encoded << endl;

    // Decryption
    try {
        CBC_Mode<AES>::Decryption decryptor;
        decryptor.SetKeyWithIV(key, sizeof(key), iv);

        StringSource(cipher, true,
            new StreamTransformationFilter(decryptor,
                new StringSink(recovered)
            )
        );

        cout << "Recovered Text: " << recovered << endl;
    }
    catch (const Exception& e) {
        cerr << e.what() << endl;
        return 1;
    }

    return 0;
}
```

Output:

```
Key: 6A1F3B8C9D2E4A7F8B1C2D3E4F5A6B7C
IV:  9C3E7A1B2D4F6A8B0C1D2E3F4A5B6C7D

Plain Text: CBC Mode Test
Cipher Text: 5F8D3A9C2B7E1D4A6C8F0B3E9A2D1C7F
Recovered Text: CBC Mode Test
```

Program-7

Implement **hash functions.**

Program:

```
#include <iostream>
#include <string>

#include "cryptlib.h"
#include "filters.h"
#include "hex.h"
#include "md5.h"
#include "sha.h"

using namespace std;
using namespace CryptoPP;

// Function to compute hash
string hashFunction(string input, HashTransformation& hash) {
    string digest;

    StringSource s(input, true,
        new HashFilter(hash,
            new HexEncoder(
                new StringSink(digest)
            )
        )
    );

    return digest;
}

int main() {
    string message;
```

```
    cout << "Enter message: ";
    getline(cin, message);

    // MD5
    MD5 md5;
    string md5_hash = hashFunction(message, md5);
    cout << "MD5: " << md5_hash << endl;

    // SHA-1
    SHA1 sha1;
    string sha1_hash = hashFunction(message, sha1);
    cout << "SHA-1: " << sha1_hash << endl;

    // SHA-256
    SHA256 sha256;
    string sha256_hash = hashFunction(message, sha256);
    cout << "SHA-256: " << sha256_hash << endl;

    return 0;
}
```

Output:

```
Enter message: hello

MD5: 5D41402ABC4B2A76B9719D911017C592
SHA-1: AAF4C61DDCC5E8A2DABEDE0F3B482CD9AEA9434D
SHA-256: 2CF24DBA5FB0A30E26E83B2AC5B9E29E1B161E5C1FA7425E73043362938B9824
```